Bias-Variance

- Bias is the tendency to learn the wrong thing by not taking into account all the information in the data (underfitting). It is also used to denote by how much the average accuracy of the algorithm changes as input/training data changes.

- Variance is the tendency to learn random things irrespective of the real signal by fitting highly flexible models that follow the error in the data too seriously (overfitting). It is used to denote how sensitive the algorithm is to the chosen input data.

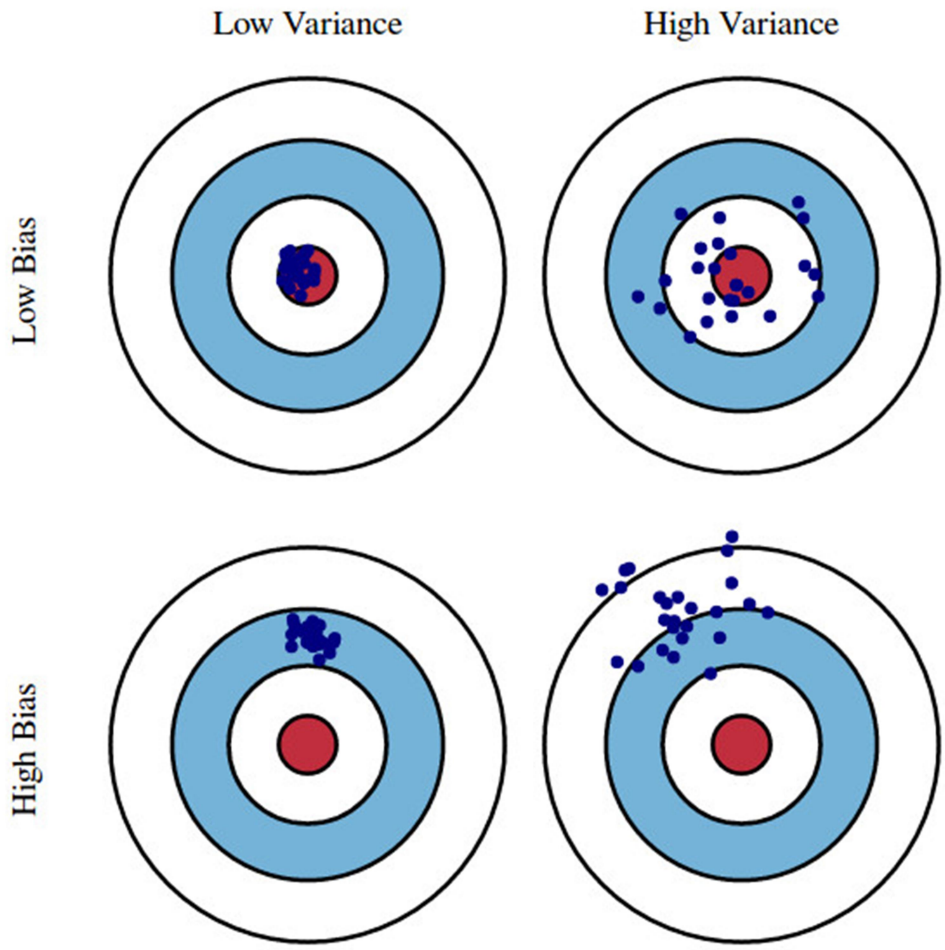

Fig. 1 Graphical illustration of bias and variance.

How does Bagging reduces the Variance?

Variance refers to how dependent the model is on the training data. If a model is highly dependent on training data (i.e. even if the training data changes slightly, the model parameters learnt will change), it is unstable and said to have a high variance.

Bagging refers to training the same model multiple times on different data sets(which are obtained by random sampling with replacement from the training set we have). All the models are combined at the end which leads to higher stability and a lower variance compared to the individual models. If all the models trained initially have a high bias, bagging cannot improve the scenario. If all models learnt initially don’t fit the training data well, they will not fit the training (or test) data when combined.

How does Boosting reduces the Bias?

- Boosting is simply training iteratively the same weak classifier, so that by weighting more to the misclassified observations. Finally, the classifier is calculated by a weighted mean/vote of all the weak classifiers.

- Usually there is a trade-off between bias and variance. A model with low bias generally have a high variance

- Boosting's objective is to increase the complexity of models that suffer from high bias especially in the case of underfitting. In simple words, boosting tries to reduce the error in predictions and hence reduces bias.